Tulio Inoue

Data Scientist | Full-Stack Developer

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PROFESSIONAL SUMMARY

Data Scientist and Full-Stack Developer with experience in data analysis, machine learning, and end-to-end software development, focusing on building scalable and data-driven solutions for real-world business challenges.

Solid experience with Python, SQL, cloud platforms, and modern web development technologies, applying techniques such as statistical analysis, NLP, machine learning, ETL pipelines, and data visualization, with a strong focus on retail economics and large volumes of transactional data.

Focused on transforming raw data into actionable insights, automating analytical workflows, and developing full-stack systems that integrate deep analytics into the business decision-making process. Actively seeking opportunities as a Data Scientist or Full-Stack Developer.

PROFESSIONAL EXPERIENCE

TOTALPASS | *Data Scientist* — Dec 2025 – Feb 2026

São Paulo, Brazil (Remote)

- Focused on the Fraud Analysis and Prevention unit.
- Developed and tested end-to-end detection algorithms, from robust data cleaning and pre-processing to the identification of outliers and anomalous patterns.

Technologies: Python, Data Science, AWS, Machine Learning.

FGV IBRE – Instituto Brasileiro de Economia | *Data Analyst* — Apr 2025 – Jul 2025

São Paulo, Brazil (On-site)

- Increased analytical coverage by over 30% through large-scale data modeling.
- Combined NLP and data engineering techniques with a focus on robustness, scalability, and economic applicability.
- Demonstrated economies of scale in over 70% of classified inputs in the civil construction sector using large-scale tax/fiscal analysis.

Technologies: Python, Data Science, NLP, FAISS, Sentence Transformers.

R-Dias Especialistas em Varejo | *Data Scientist & Full-Stack Developer* — Apr 2024 – Nov 2024

São Paulo, Brazil (On-site)

- Enabled faster, data-driven decisions by delivering analytical dashboards and structured data pipelines.
- Boosted operational efficiency by over 40% through the automation of critical analytical workflows.
- Led the creation and development of a store expansion forecasting web application, supporting key strategic choices and scaling previous projects by more than 2x.

Technologies: Python, Pandas, Matplotlib, SQL, Machine Learning, Full-Stack Development.

R-Dias Especialistas em Varejo | *Statistical Analyst* Sep 2023 – Apr 2024

São Paulo, Brazil (On-site)

- Reduced data processing time by 3 hours through efficient pipeline and data flow automation.
- Increased analytical effectiveness by over 25% by delivering an interactive executable tool for advanced statistical visualization.
- Built dynamic dashboards and visualizations to align strategic goals and support technical discussions with key stakeholders.

Technologies: Python, Pandas, Tkinter, Statistical Analysis, Data Visualization, Power BI.

EDUCATION

Universidade de São Paulo (USP) Feb 2018 – Jun 2022

Bachelor's Degree in Mathematics (Licenciatura)

Fundação de Negócios, Analytics e Tecnologia (FNAT) Feb 2026 – Dec 2026

Postgraduate Specialization in Data Analysis and Artificial Intelligence Applied to Business

PROJECTS

Diabetes Prediction and Classification

- Applied robust data pre-processing, resampling techniques, and extensive performance evaluation.
- Developed a machine learning model for diabetes risk prediction and classification leveraging WHO medical databases.
- Deployed the production-ready model inside the AWS cloud ecosystem.
- Created and deployed a React + TypeScript frontend via S3, integrating seamlessly with the Lambda-hosted model backend.

Technologies: Python, scikit-learn, Matplotlib, NumPy, Seaborn, Pandas, Polars, Joblib, HTML, CSS, React, TypeScript, echarts-for-react, AWS Lambda, API Gateway, S3, GitHub.

Real-Time Atmospheric Dashboard for Lung Health

- Designed an ETL pipeline with scheduled continuous data collection, archived in Amazon DynamoDB (NoSQL).
- Built and deployed a frontend dashboard using React + TypeScript via S3, connecting securely to the database through AWS Lambda.

Technologies: Python, HTML, CSS, React, TypeScript, echarts-for-react, AWS Lambda, EventBridge, DynamoDB, API Gateway, S3, GitHub.

Lung Pathology Segmentation and Classification

- Application of computer vision pipelines, U-Net anatomical segmentation, and multi-pathology classification with Deep Learning.
- Development of a cascade architecture (Model 1: Lung segmentation via U-Net; Model 2: Classification and diagnosis of COVID-19, Lung Opacity, and Pneumonias via EfficientNetB2).
- Deployed the production-ready model inside the AWS cloud ecosystem on ONNX format.
- Creation and S3 deployment of a React + TypeScript front-end with asynchronous connections to the models implemented in a Serverless environment (Lambda and API Gateway).

Technologies: Python, scikit-learn, Matplotlib, NumPy, Seaborn, Pandas, Polars, Joblib, HTML, CSS, React, TypeScript, echarts-for-react, AWS Lambda, API Gateway, S3, GitHub.

SKILLS

Data Science & Analytics: Python, SQL, Statistics, Machine Learning, NLP, ETL, Data Visualization, Pandas, NumPy, Scikit-learn.

Cloud & Tools: AWS (Lambda, API Gateway, S3, DynamoDB, EventBridge), GitHub, Git.

Web & Full-Stack: HTML, CSS, JavaScript, React, TypeScript, Node.js.

BI & Visualization: Power BI, Excel, Streamlit, Matplotlib, Seaborn.

LANGUAGES

Portuguese: Native | **English:** Advanced | **Spanish:** Intermediate